Parity of fixed points in triangular-divisor sequences

A solution to MathDB 343324

Principia Math In Progress

18 September 2026

Abstract

We prove the parity conjecture for the sequences of Navarrete and Orellana. The obstruction is already present in the defining divisibility condition: at an even index, the index does not divide the prescribed odd multiple of a triangular number. Primality and the greedy selection rule are unnecessary for this obstruction. For comparison, every even positive multiplier gives the identity sequence under the greedy rule.

Task 1: the original problem and its full solution

For an odd prime p, put

$$q_p(n) = \sum_{j=0}^{n-1} jp = \frac{pn(n-1)}{2}.$$

The source [1] defines $a_p(1) = 1$ and, for $n \geq 2$, chooses $a_p(n)$ to be the least positive divisor of $q_p(n)$ not among $a_p(1), \ldots, a_p(n-1)$. Its Conjecture 3.1, on page 4, asserts that $a_p(n) = n$ can occur only at odd n.

Theorem 1. *Let m be an odd positive integer. If n is a positive integer and n divides $mn(n-1)/2$, then n is odd. Consequently, every fixed point of $A(p)$ is odd for every odd prime p.*

Proof. The divisibility assumption says that, for some integer b,

$$\frac{mn(n-1)}{2} = nb.$$

Since $n > 0$, cancellation gives $m(n-1) = 2b$. The integer m is odd, so $n-1$ is even and n is odd. For the application, the defining rule always gives $a_p(n) \mid q_p(n)$. Substituting $a_p(n) = n$ supplies exactly the preceding divisibility assumption. This also includes $n = 1$, since every nonzero integer divides zero. □

An equivalent explicit remainder is useful for checking the obstruction. Writing $m = 2\ell + 1$ and $n = 2k$, where $\ell \geq 0$ and $k \geq 1$, gives

$$\frac{mn(n-1)}{2} = 2k\big((2\ell+1)k - \ell - 1\big) + k \equiv k \pmod{2k}. \tag{1}$$

The remainder is strictly between zero and the modulus. No information about earlier sequence entries is required.

Corollary 1. *For each integer $m \geq 2$, define the greedy triangular-divisor sequence with multiplier m by the same rule. It is well defined. If m is odd, all its fixed points are odd. If m is even, then $a_m(n) = n$ for every $n \geq 1$.*

Proof. For $n \geq 2$, the positive numbers $q_m(n)$ strictly increase and $q_m(2) = m > 1$. Every previous selected divisor is at most its corresponding $q_m(j)$, apart from the initial value 1. Thus $q_m(n)$ itself has not been selected and is an available divisor. This proves existence at each step. The odd case follows from Theorem 1. If m is even, then $q_m(n) = n\big((m/2)(n-1)\big)$, so n is an available divisor whenever the preceding values are $1, \ldots, n-1$. Every smaller positive integer has already been selected; therefore the greedy rule selects n. Induction starts at $a_m(1) = 1$. □

Scope and attribution. This resolves the original all-prime parity statement, without changing its indexing. The result does not say that odd indices are fixed points or that fixed points are prime. The source's Table 4 already records the identity sequence for multipliers $2, 4, 6$; Corollary 1 includes these observations. The later paper [2] proves a separate result on prime fixed points below $2p$, along with other assertions. Its results are credited, and are not used in the proof above.

Task 2: dependence on odd composite multipliers

For odd composite m, determine when the complete greedy sequence $A(m)$ agrees with $A(p)$ for a prime divisor p of m. The source reports agreement for $m = 9$ and $p = 3$. Establish a criterion or locate the first disagreement for other prime powers and products of distinct primes.

Task 3: distribution of composite fixed points

For a fixed odd prime p, study

$$C_p(X) = \#\{n \leq X : n \text{ is composite and } a_p(n) = n\}.$$

Determine whether this count has a natural density and obtain bounds that use the exclusion of even indices. The existence of composite fixed points is already documented in the sources; their distribution is the proposed question.

Task 4: prime fixed points beyond the known range

Theorem 5.1 of [2] gives $a_p(\ell) = \ell$ for odd primes $\ell \neq p$ with $\ell < 2p$. Determine which additional primes outside this range are fixed points. Seek a criterion involving the prior availability of divisors of $p\ell(\ell-1)/2$ and quantitative bounds on the exceptional primes.

Task 5: other integer-valued input polynomials

Replace $mn(n-1)/2$ by another increasing integer-valued polynomial $Q(n)$ and choose the least unused positive divisor at each step. Classify the residue classes forbidden to fixed points by the necessary condition $n \mid Q(n)$. Then determine when these local obstructions account for all missing fixed-point classes.

Verification and status. The proof is elementary and does not depend on computation. The accompanying script checks the exact remainder identity, reproduces an independently recorded sequence, and checks finite prefixes for odd and even multipliers. The literature audit documents the current primary, the related 2022 paper, a bounded prior-claim search, and live MathDB checks. It is not an exhaustive novelty guarantee. Review was by the authoring agent only.

References

[1] E. Navarrete and D. Orellana, *Finding Prime Numbers as Fixed Points of Sequences*, arXiv:1907.10023v1 (2019), 9 pages. arXiv record.

[2] J. A. Julca Avila and G. Silva de Andrade, *About the theorem that partially solves the Navarrete–Orellana Conjecture*, Notes on Number Theory and Discrete Mathematics **28**(2) (2022), 302–317. doi:10.7546/nntdm.2022.28.2.302-317.